

CHECK PTH FIRST THEN PTHrP THEN 1,25 D

PRIMARY HYPER PTH
MEN1 MEN1 MEN2A
CDC73 JAW TUMOR
FHH
FE Ca LESS 1 PERCENT

HIGH PTH
CHECK PTH
HIGH SERUM Ca

CHECK PTHrP
MALIGNANCY HUMORAL SQUAMOUS BREAST
HIGH PTHrP
LOW PTHrP
CHECK 1,25-D
HIGH 1,25-D
GRANULOMATOUS MACROPHAGE 1 ALPHA HYDROXYLASE
LOW 1,25-D
OTHER VIT D INTOX PAGES IMMOBIL MILK ALKALI

VITAMINS TRAP

CALCIUM SEVERITY
MILD 10.5 to 11.9
MODERATE 12 to 14
SEVERE OVER 14

HUNGRY BONE SYNDROME after PARATHYROIDECTOMY
CALCIUM INTO HIS BONES

ALKALOSIS INCREASES Ca BINDING TO ALBUMIN BOTH RESPIRATORY AND METABOLIC

1. PTH → PTHrP → 1,25-D

WHAT YOU SEE

Banner: CHECK PTH FIRST THEN PTHrP THEN 1,25 D

WHAT IT ENCODES

The single most important board reflex for any hypercalcemia: measure PTH FIRST. PTH cleanly splits the differential into PTH-mediated (high/inappropriately-normal) vs PTH-independent (suppressed). Only if PTH is suppressed do you proceed to PTHrP (humoral malignancy) and then 1,25-OH vitamin D (granulomatous/lymphoma). Roughly 90% of all hypercalcemia is either primary hyperparathyroidism or malignancy, and PTH distinguishes them in one test.

BOARD TRAP

WRONG: order a serum calcium-only panel or jump to a parathyroid sestamibi scan before checking PTH. Imaging localizes an adenoma but never establishes the diagnosis — PTH is the branch-point. Ionized calcium confirms true hypercalcemia but does not tell you the cause.

2. High serum Ca trunk

WHAT YOU SEE

Tree base: HIGH SERUM Ca

WHAT IT ENCODES

Confirmed high serum calcium is the trunk that splits into two branches. Before working it up, confirm it is REAL: correct for albumin (corrected Ca = measured + 0.8 × [4 – albumin]) or get an ionized calcium, since dehydration and high albumin/paraprotein can spuriously raise total calcium. Normal total Ca is ~8.5–10.5 mg/dL.

BOARD TRAP

WRONG: treating an elevated TOTAL calcium in a patient with high albumin (hemoconcentration) or a paraprotein (myeloma) as true hypercalcemia. Ionized calcium is normal in pseudohypercalcemia — no workup or treatment needed.

3. High PTH branch

WHAT YOU SEE

Left branch: HIGH PTH, CHECK PTH

WHAT IT ENCODES

A high OR inappropriately-normal PTH in the face of hypercalcemia = PTH-mediated disease. Normally high calcium should suppress PTH via the CaSR, so any non-suppressed PTH is pathologic. This branch holds primary (and tertiary) hyperparathyroidism and FHH. Confirm with 24-hour urine calcium to separate FHH (low) from primary hyperPTH (high/normal).

BOARD TRAP

WRONG: calling a PTH value that sits in the normal range 'normal.' With elevated calcium, a mid-normal PTH is INAPPROPRIATELY normal and still indicates PTH-mediated hypercalcemia.

4. Suppressed PTH branch

WHAT YOU SEE

Right branch: SUPPRESSED PTH, CHECK PTHrP

WHAT IT ENCODES

A low/suppressed PTH means the parathyroids are appropriately responding — the calcium is being driven from elsewhere (PTH-independent). Next steps: PTHrP (humoral hypercalcemia of malignancy), 1,25-D (granulomatous disease/lymphoma), 25-D (vitamin D intoxication), and SPEP/imaging for myeloma or lytic mets. Malignancy dominates this branch in inpatients.

BOARD TRAP

WRONG: ordering a PTHrP level FIRST in every hypercalcemia. PTHrP is only useful once PTH is confirmed suppressed; checking it up front wastes a send-out test and confuses the algorithm.

5. Primary hyperPTH

WHAT YOU SEE

A little parathyroid-gland creature labelled PRIMARY HYPER PTH perched on the high-PTH branch — it sits on the high-PTH side because an autonomous gland keeps over-secreting PTH.

WHAT IT ENCODES

Primary hyperparathyroidism is the #1 OUTPATIENT cause — usually a single autonomous adenoma (~85%), often found incidentally as mild asymptomatic hypercalcemia with high/normal PTH and low phosphate. Surgical cure (parathyroidectomy) is indicated for Ca >1 mg/dL above normal, age <50, osteoporosis/fragility fracture, or eGFR <60/nephrolithiasis. Consider MEN1 and MEN2A in young/multigland disease.

BOARD TRAP

WRONG: assuming the most common cause of hypercalcemia OVERALL is also the most common in a sick hospitalized patient. Inpatient hypercalcemia is malignancy until proven otherwise — primary hyperPTH is the outpatient/incidental winner.

6. CDC73 / jaw tumor

WHAT YOU SEE

A branch tag reading CDC73 / JAW TUMOR with a figure clutching its jaw — the jaw mass flags the CDC73 (parafibromin) syndrome and its parathyroid-carcinoma risk.

WHAT IT ENCODES

Hyperparathyroidism–jaw tumor syndrome is caused by germline CDC73 (HRPT2, parafibromin) mutations. It carries a notably HIGH risk of parathyroid CARCINOMA (markedly elevated Ca and very high PTH) plus ossifying fibromas of the jaw and renal lesions. Think of it in a young patient with severe hyperPTH and a jaw mass.

BOARD TRAP

WRONG: lumping it with sporadic adenoma. A young patient with very high PTH, markedly elevated calcium, and a palpable neck mass should raise parathyroid carcinoma / CDC73, not benign adenoma.

7. FHH

WHAT YOU SEE

A branch labelled FHH next to a kidney holding onto calcium — the kidney avidly reabsorbing Ca (low urine calcium) is the visual signature of familial hypocalciuric hypercalcemia.

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia is an autosomal-dominant INACTIVATING CaSR mutation: the receptor under-senses calcium, so PTH and Ca run mildly high but the kidney avidly reabsorbs calcium → LOW urine calcium. The discriminator is the fractional excretion of calcium <0.01 (1%); 24-h urine Ca is low. It is benign and lifelong — no surgery, no treatment.

BOARD TRAP

WRONG: sending a patient with high Ca and high/normal PTH to parathyroidectomy without checking urine calcium. Operating on FHH does nothing (hypercalcemia persists) — the FECa <1% (vs >2% in primary hyperPTH) prevents the needless surgery.

8. Malignancy PTHrP

WHAT YOU SEE

Right top: MALIGNANCY HUMORAL, squamous, breast

WHAT IT ENCODES

Humoral hypercalcemia of malignancy is the #1 INPATIENT cause: tumor-secreted PTHrP mimics PTH (binds the PTH receptor) → high Ca, LOW phosphate, but SUPPRESSED native PTH. Classic primaries: squamous cell (lung, head/neck), renal, bladder, and breast carcinoma. It signals advanced disease and poor prognosis; treat the calcium AND the tumor.

BOARD TRAP

WRONG: expecting PTH to be HIGH because PTHrP 'acts like PTH.' The assay measures intact PTH, which is appropriately SUPPRESSED — high calcium with low PTH and low phosphate plus a known cancer = PTHrP-mediated.

9. Check PTHrP node

WHAT YOU SEE

A wooden trail sign on the suppressed-PTH branch reading CHECK PTHrP — the sign points downstream because PTHrP is the SECOND test, ordered only after PTH comes back low.

WHAT IT ENCODES

This is the second-tier test on the suppressed-PTH branch. After PTH comes back low, an elevated PTHrP nails humoral hypercalcemia of malignancy. If PTHrP is also normal/low, pivot to 1,25-D (granulomatous/lymphoma), 25-D (vitamin D excess), SPEP/UPEP and skeletal survey (myeloma), and TSH/cortisol for endocrine mimics.

BOARD TRAP

WRONG: stopping the workup at a normal PTHrP and declaring no malignancy. Lytic bone metastases and myeloma cause hypercalcemia via local cytokines (RANKL/IL-6), NOT PTHrP — PTHrP can be normal while cancer is the cause.

10. Granulomatous 1,25-D

WHAT YOU SEE

Granulomatous, macrophage 1-alpha-hydroxylase, high 1,25-D

WHAT IT ENCODES

In sarcoidosis, TB, and lymphoma, activated macrophages express UNREGULATED 1-alpha-hydroxylase, converting 25-D to active 1,25-D independent of PTH. Result: high 1,25-D, suppressed PTH, high calcium and often hypercalciuria. It responds dramatically to GLUCOCORTICOIDS (prednisone), which shut down macrophage 1-alpha-hydroxylase.

BOARD TRAP

WRONG: treating granulomatous hypercalcemia primarily with bisphosphonates or heavy sun exposure/vitamin D. Steroids are the targeted therapy, and these patients should AVOID vitamin D and sunlight, which fuel the conversion.

11. Low 1,25-D group

WHAT YOU SEE

LOW 1,25-D, other vit D intox, Paget, immobilization, milk-alkali

WHAT IT ENCODES

PTH-independent causes with LOW or normal 1,25-D: vitamin D INTOXICATION raises 25-D (not 1,25-D), Paget disease + immobilization releases skeletal calcium, and milk-alkali syndrome (calcium + absorbable alkali) gives the triad of hypercalcemia, metabolic alkalosis, and AKI. Thiazides and lithium also live on the drug side of this branch.

BOARD TRAP

WRONG: ordering 1,25-D to diagnose vitamin D TOXICITY. Exogenous vitamin D excess raises 25-OH-D (the storage form measured for toxicity); 1,25-D is often normal/low. Check 25-D for intoxication, 1,25-D for granulomatous disease.

12. Calcium severity scale

WHAT YOU SEE

Mild 10.5-11.9, moderate 12-14, severe >14

WHAT IT ENCODES

Severity guides urgency: MILD 10.5–11.9 (often asymptomatic, outpatient workup), MODERATE 12–14 (symptomatic, treat), SEVERE >14 mg/dL or symptomatic = hypercalcemic emergency needing aggressive IV saline + calcitonin + bisphosphonate. Above ~14, neurologic changes and arrhythmia risk rise sharply.

BOARD TRAP

WRONG: discharging a patient with calcium >14 because they 'look okay.' Severe hypercalcemia is an emergency regardless of how mild symptoms seem — treat immediately, don't wait for the workup.

13. Hungry bone syndrome

WHAT YOU SEE

Kneeling figure: HUNGRY BONE after parathyroidectomy

WHAT IT ENCODES

After parathyroidectomy for severe/long-standing hyperPTH, the abrupt drop in PTH lets demineralized bone avidly reabsorb minerals → profound, prolonged HYPOcalcemia, hypophosphatemia, and hypomagnesemia. It is distinguished from transient post-op hypoparathyroidism by LOW phosphate (hypoPTH gives HIGH phosphate). Treat with aggressive calcium + calcitriol.

BOARD TRAP

WRONG: confusing hungry bone with surgical hypoparathyroidism. Both give post-op hypocalcemia, but hungry bone has LOW phosphate and elevated/normal PTH, whereas hypoparathyroidism has HIGH phosphate and low PTH — the phosphate level discriminates.

14. Vitamins / alkalosis trap

WHAT YOU SEE

VITAMINS TRAP sign: alkalosis increases Ca binding to albumin

WHAT IT ENCODES

Alkalosis (respiratory or metabolic) increases calcium binding to albumin, dropping IONIZED (active) calcium while total calcium stays the same — this is why acute hyperventilation can trigger perioral tingling and carpopedal spasm with a 'normal' total Ca. Conversely, acidosis frees ionized calcium. Always interpret calcium alongside pH and albumin.

BOARD TRAP

WRONG: reassuring a hyperventilating patient with tetany because total calcium is normal. The ionized fraction has fallen due to alkalosis — symptomatic hypocalcemia can exist with a perfectly normal total calcium.
